

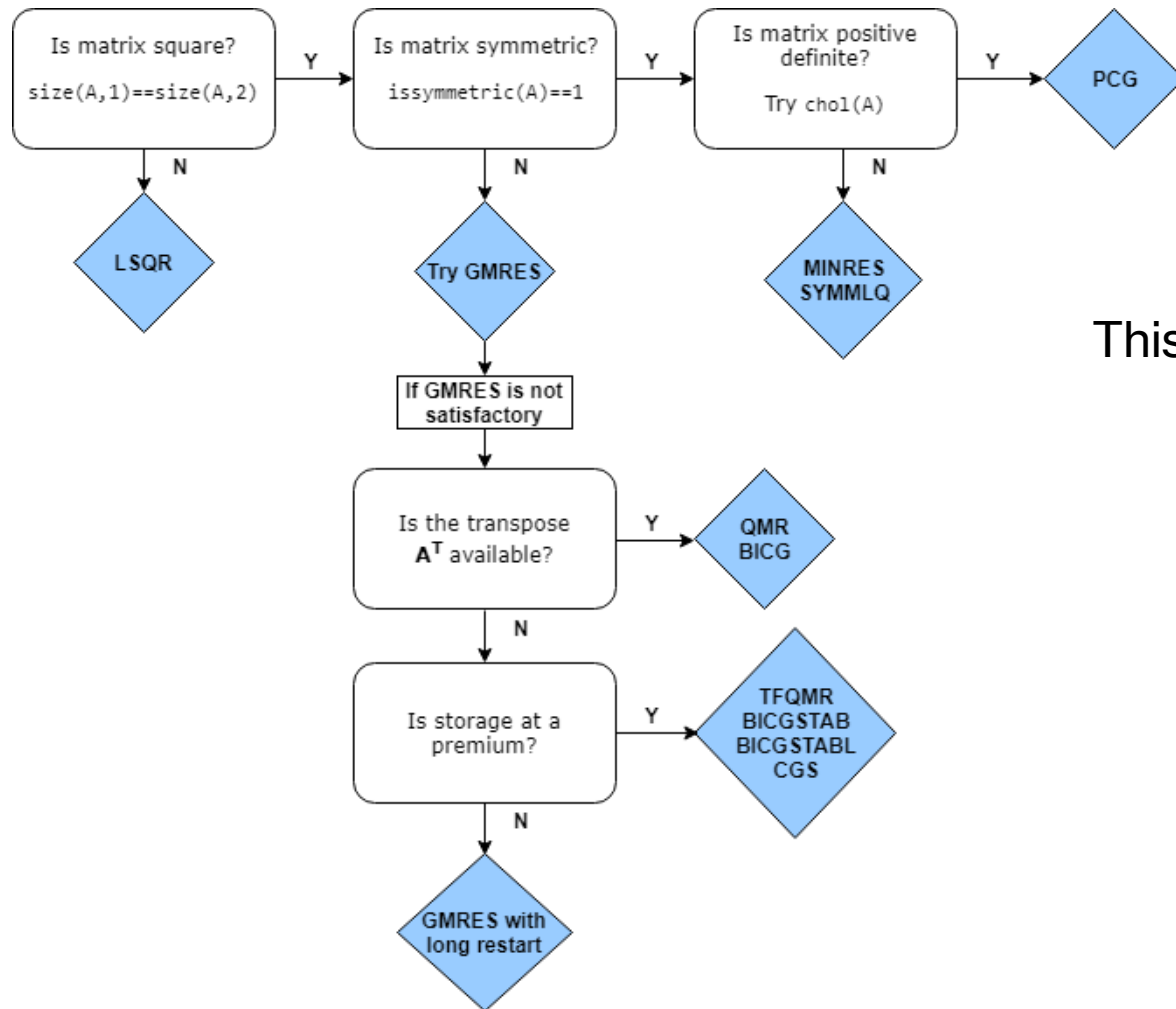
The matrix-vector complexity of $Ax=b$

Information-theoretic hardness results.

Presented by Raphael Meyer

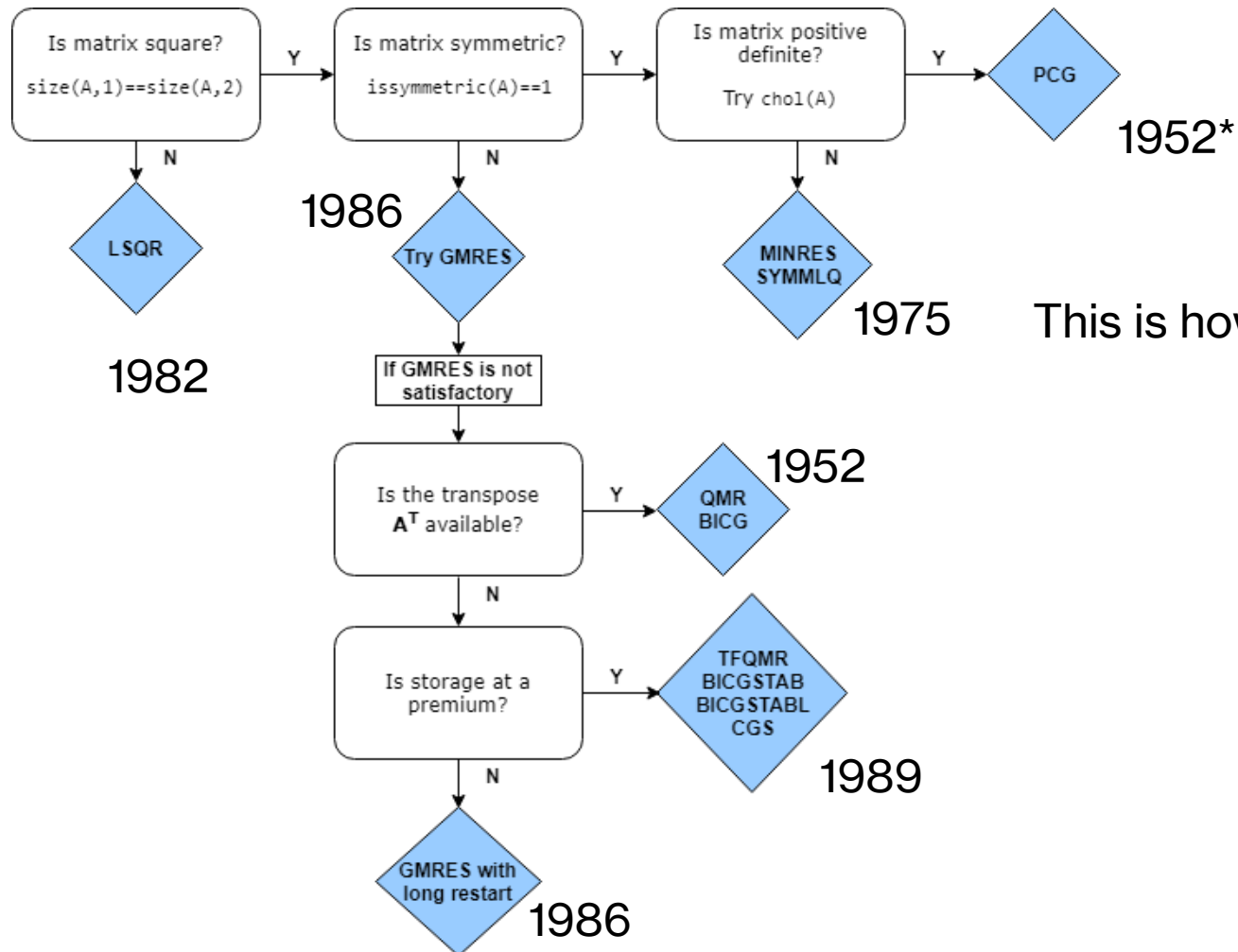
Joint work with Ethan Epperly and Michal Dereziński. Accepted to COLT 2026.

MATLAB's $Ax=b$ solvers of choice.



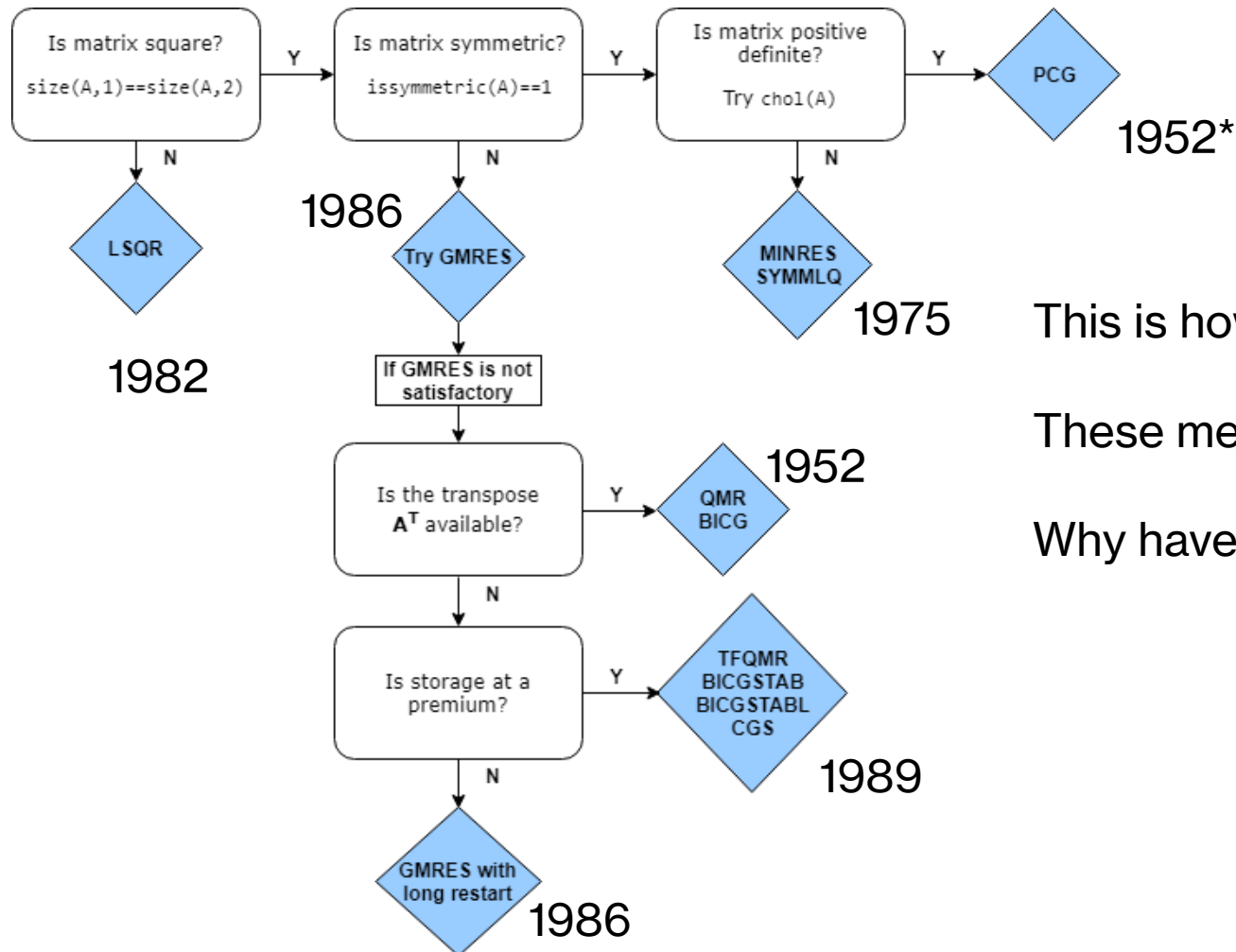
This is how MATLAB solves a **square** linear system

MATLAB's $Ax=b$ solvers of choice.



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These methods are between 35 and 75 years old

Why have they not changed?

Are these algorithms “optimal”?

CG	CG on $A^\top Ax = A^\top b$ <i>(or MINRES)</i>	GMRES
A is SPSPD	Arbitrary A	Arbitrary A
$O(\sqrt{\kappa} \log 1/\varepsilon)$ Iterations	$O(\kappa \log 1/\varepsilon)$ Iterations	n Iterations
Computes $x \rightarrow Ax$	Computes $x \rightarrow Ax$ and $x \rightarrow A^\top x$	Computes $x \rightarrow Ax$

In what way can we say these methods are “optimal”?

Are they optimal in a RandNLA world?

Matrix-Vector Complexity



Matrix-Vector Oracle Model

View the matrix-vector product as our computational atom

$$x \rightarrow \text{black box} \rightarrow Ax$$

- This is the only way we interact with A
- CG/MINRES/GMRES/... all have this shape

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Problem Statement

Given $A \in \mathbb{R}^{n \times n}$, $\mathbf{b} \in \mathbb{R}^n$, $\varepsilon > 0$, return $\tilde{\mathbf{x}} \in \mathbb{R}^n$ s.t.
$$\|\tilde{\mathbf{x}} - A^{-1}\mathbf{b}\|_2 \leq \varepsilon \|A^{-1}\|_2 \|\mathbf{b}\|_2$$

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- What is the fewest possible number of matrix-vector products needed to solve this for all $\text{cond}(A) \leq \kappa$?
- Methods matching this have “optimal matvec complexity”

the conjugate gradients method. Then:

$$(4.2) \quad \|Ag - f\| \leq \left(\sum_{j=1}^{k+1} \left| \frac{\prod_{i \neq j} \mu_i}{\prod_{i \neq j} (\mu_j - \mu_i)} \right| \right)^{-1} \|f\|,$$

where μ_i are the eigenvalues of B .

If we define $T(x)$ to be the polynomial that has the least deviation from 0 on the interval $[\mu_{\min}, \mu_{\max}]$ and satisfies $T(0) = -1$ we can use it to get a simpler estimate than (4.2). If the deviation from zero is ϵ , then $|T(\mu_i)| \leq \epsilon$ and we may apply Theorem 2.2. It is well known that:

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Theorem: Any (possibly randomized) algorithm that converges must compute $\gtrsim \kappa \log 1/\epsilon$ matvecs.

- Constants matches upper bound to a factor of 4!
- Randomness does not help!
- There are reasons to not run CG on the normal equations.

These methods assume we can
compute Ax and $A^T x$.

What if we can only do Ax ?

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What if we can only do Ax ?

Enter: Transpose-Free Numerical Linear Algebra

Transpose-Free Methods

Some reasons why you don't want to compute $x \mapsto Ax$

Transpose-Free Methods

Some reasons why you don't want to compute $x \mapsto Ax$

- It's slower because of hardware
- It's noisier when $A^T x$ is a gradient
- More reasons...

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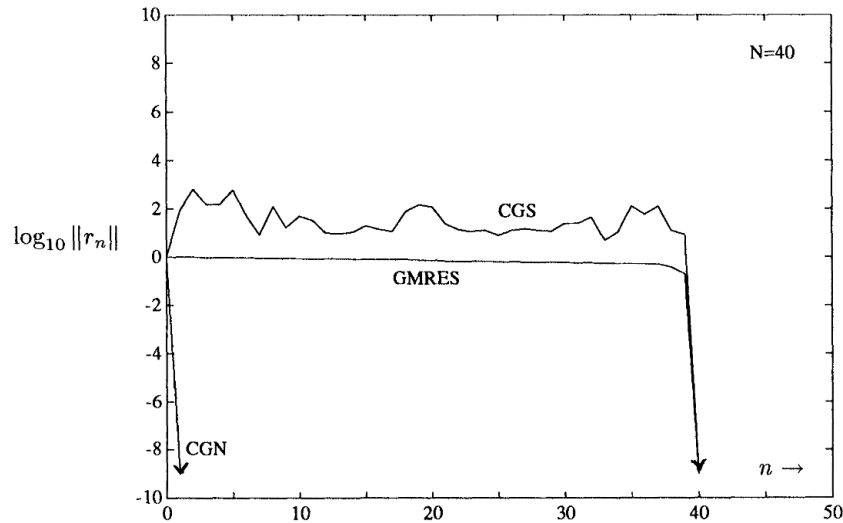
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A THEORETICAL COMPARISON OF THE ARNOLDI
AND GMRES ALGORITHMS*

PETER N. BROWN†

HOW FAST ARE NONSYMMETRIC MATRIX ITERATIONS?*

NOËL M. NACHTIGAL†, SATISH C. REDDY‡, AND LLOYD N. TREFETHEN§



“Example C” from Nachtigal et al. Transpose-free algorithms GMRES and CGS both need n matvecs.

Transpose-Free Methods

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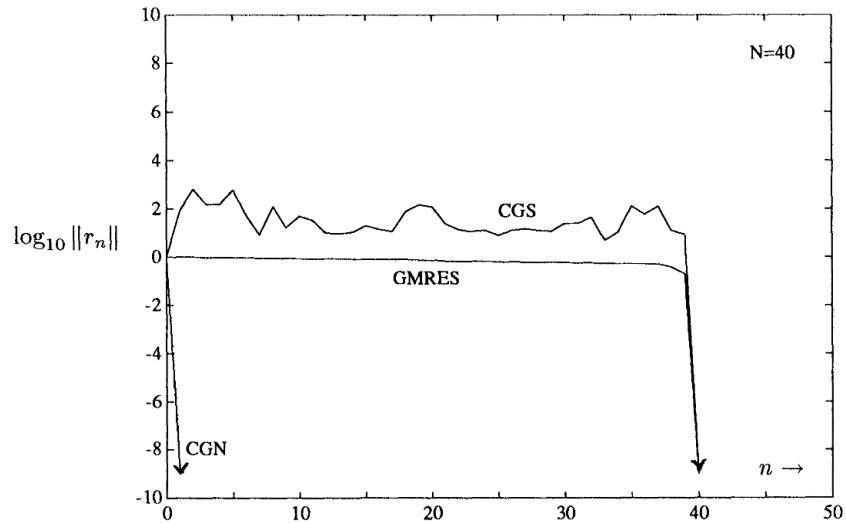
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A THEORETICAL COMPARISON OF THE ARNOLDI
AND GMRES ALGORITHMS*

PETER N. BROWN†

HOW FAST ARE NONSYMMETRIC MATRIX ITERATIONS?*

NOËL M. NACHTIGAL†, SATISH C. REDDY‡, AND LLOYD N. TREFETHEN§



“Example C” from Nachtigal et al. Transpose-free algorithms GMRES and CGS both need n matvecs.

Transpose-Free Methods

Some reasons why you don’t want to compute $x \mapsto Ax$

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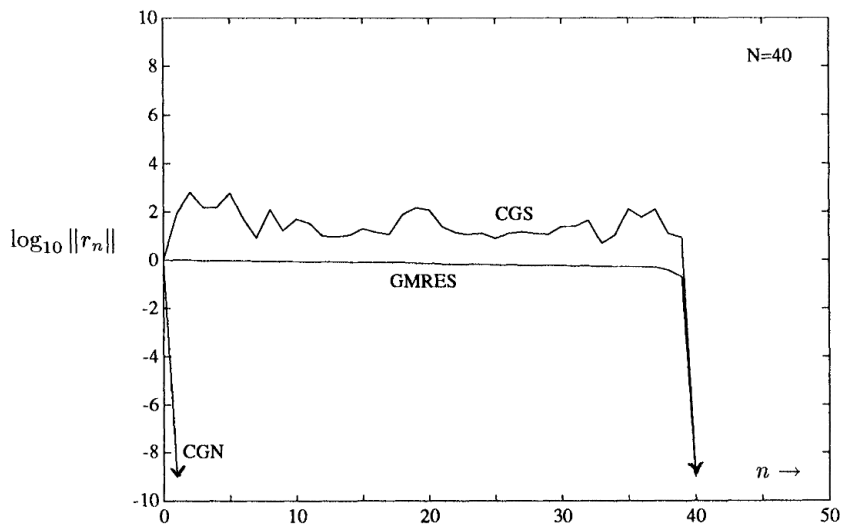
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- Randomness does not help!
- Machine precision isn’t possible! (*if $n < 10^{16}$*)
- This is worst case analysis: GMRES clearly works

Key Implication

Faster $Ax=b$ solvers must do one of three things:

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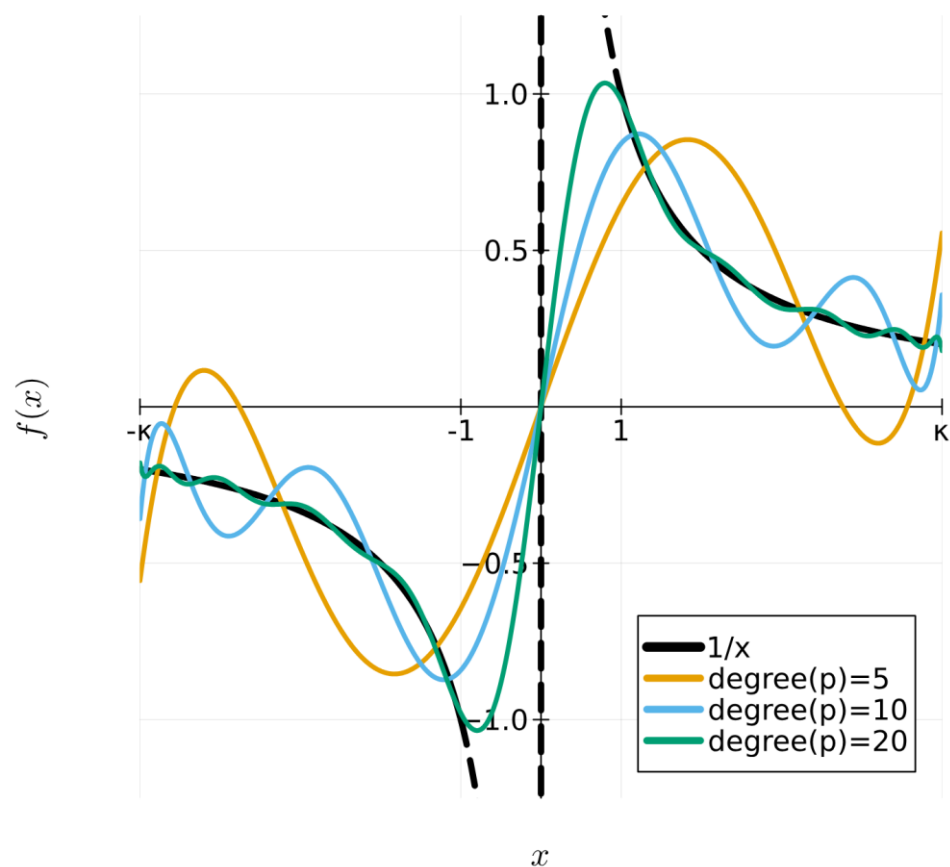
Next: How to prove the theorems.

Polynomial Approximability and Matvec Complexity

Warmup: Upper Bound

Why does MINRES/CGNR converge in $O(\kappa \log 1/\varepsilon)$?

Poly. Approx. to $1/x$ on $[-\kappa, -1] \cup [1, \kappa]$

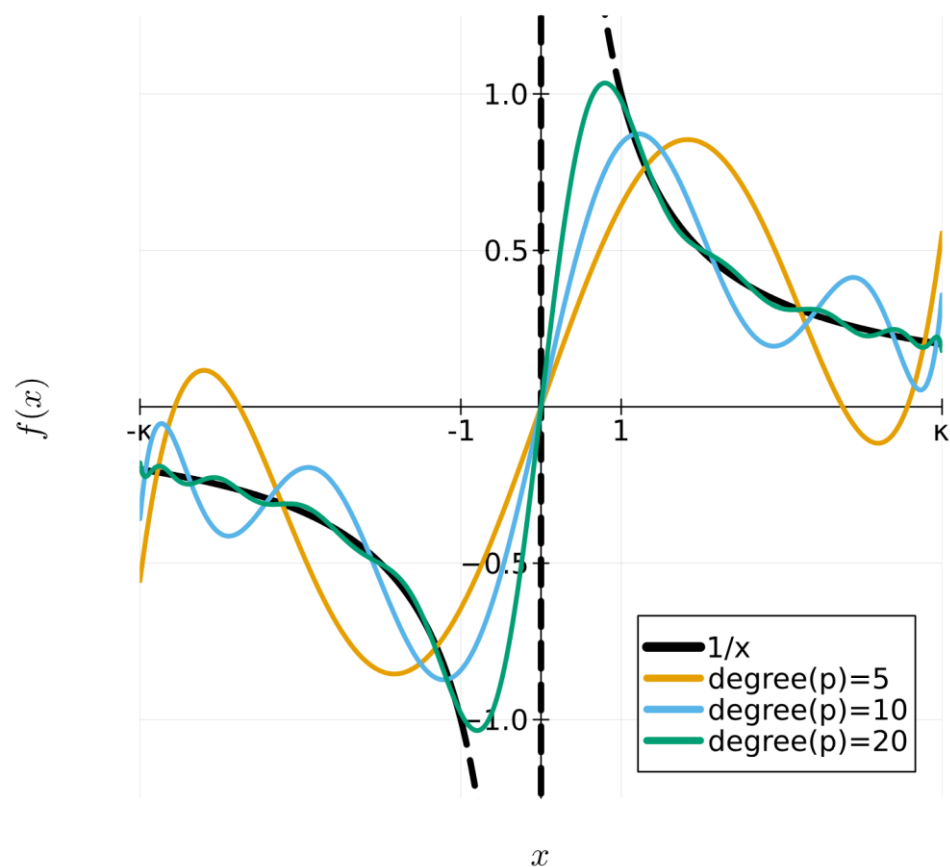


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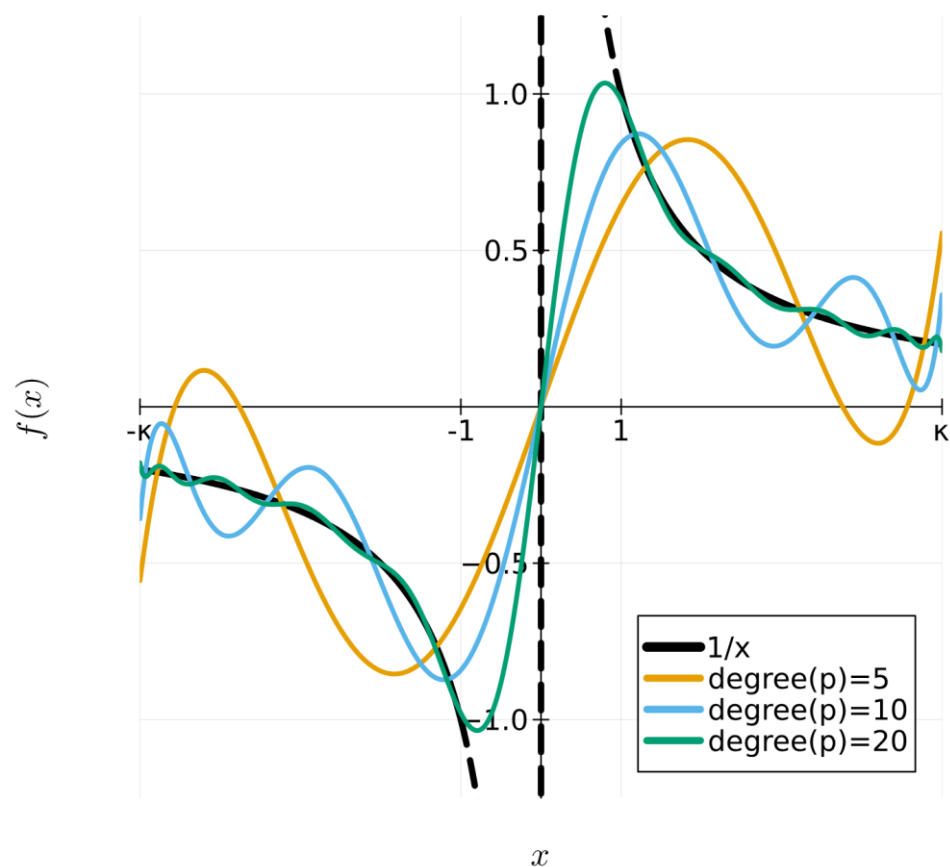
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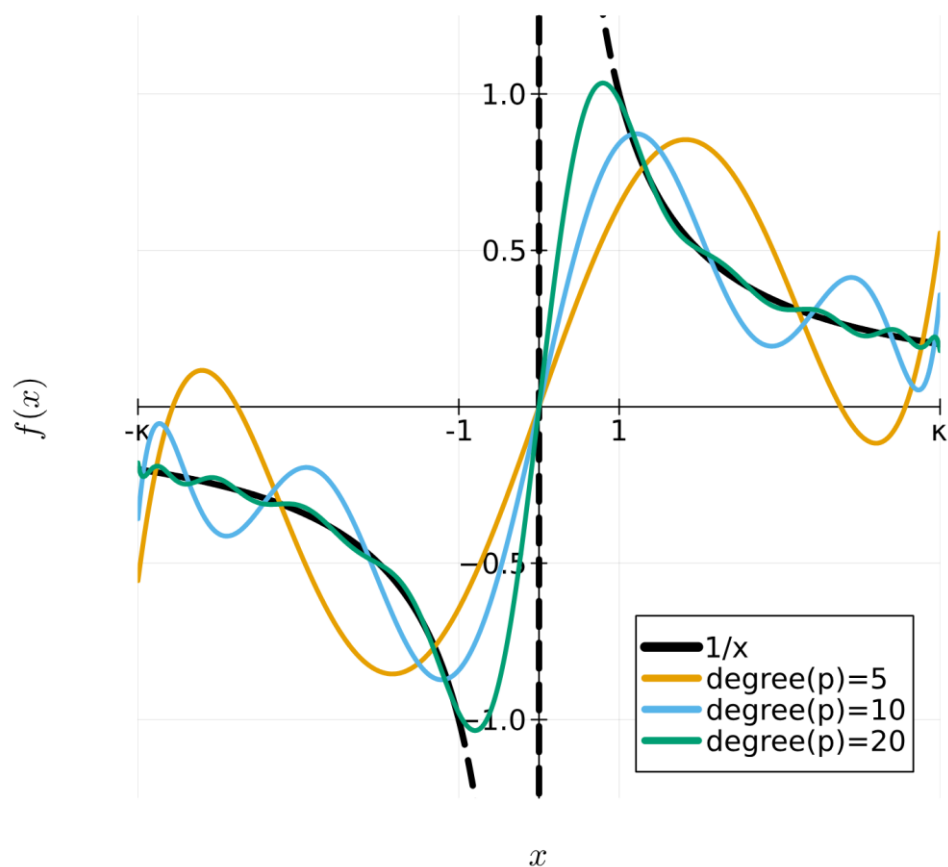
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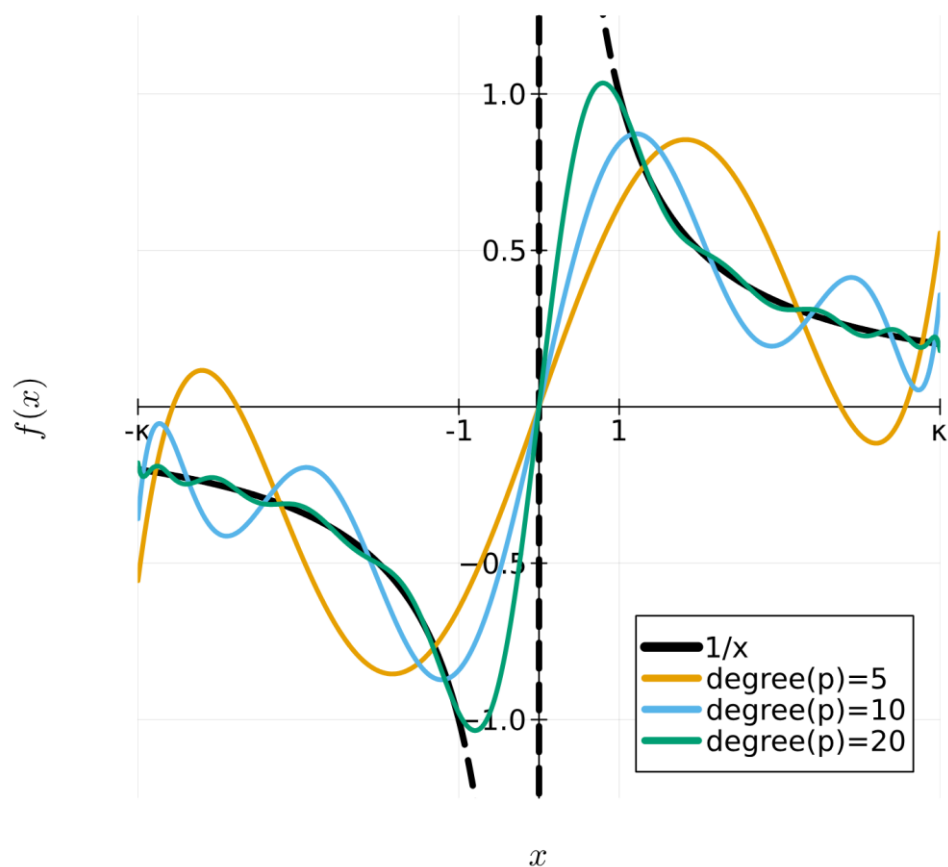
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There exists p of degree t with

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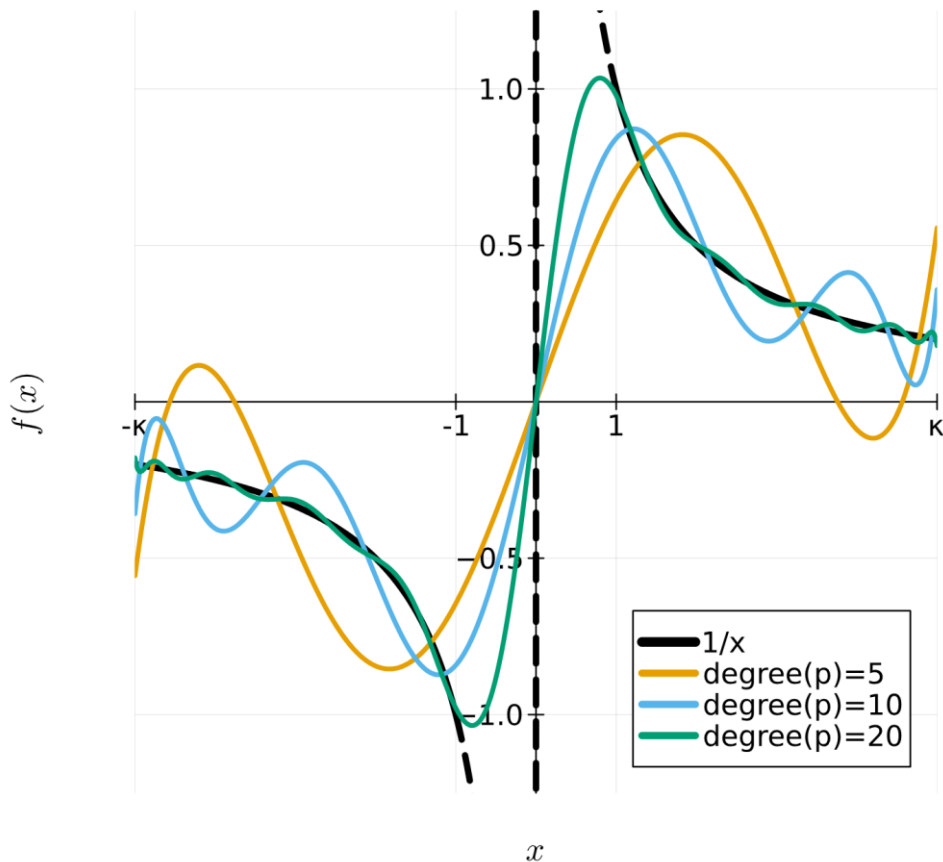
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Idea: Flip this perspective into a lower bound!

Lower bound

Any linear system solver uses at least $\Omega(\kappa \log(1/\varepsilon))$ matvecs

3 step proof! (15 pages)

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Any polynomial that has $|p(x) - x^{-1}| < \varepsilon$ for all $x \in \mathcal{D}_\kappa$ must have degree $t_\star \gtrsim \kappa \log(1/\varepsilon)$.

Basic Polynomial Theory

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Any algorithm that estimates $\text{tr}(A^{-1})$ to error $\|A^{-1}\|_F$ must compute $\gtrsim t_\star$ matvecs.

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Girard-Hutchinson Estimator

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In the paper:

- Good constants
- Different error metrics
- SPSP Case
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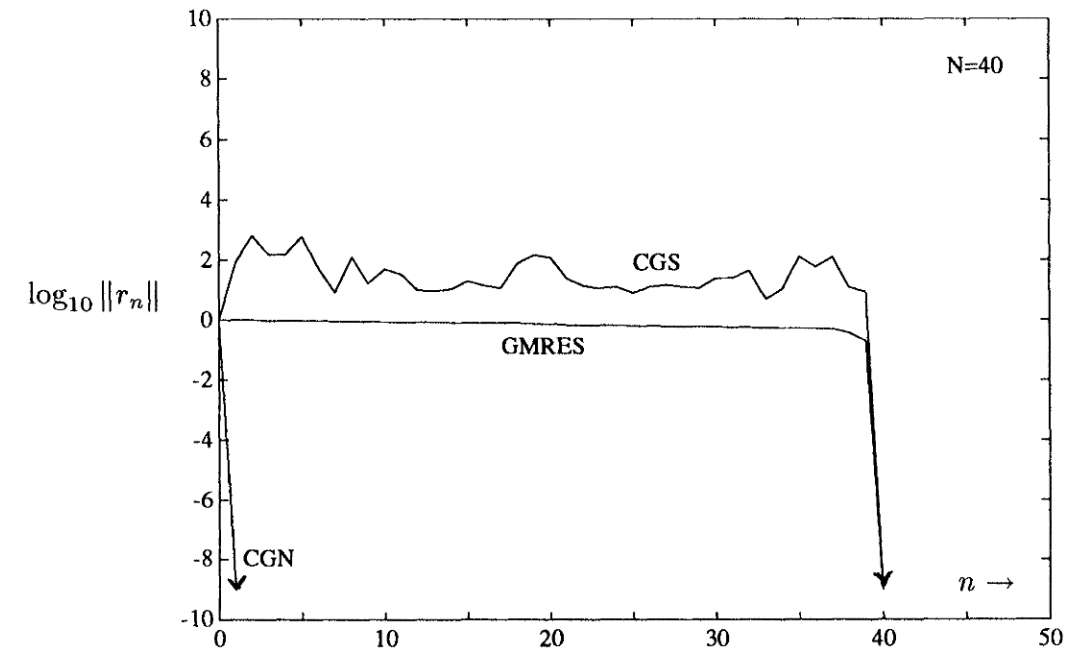
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Random Orthogonal Matrices and Matvec Complexity

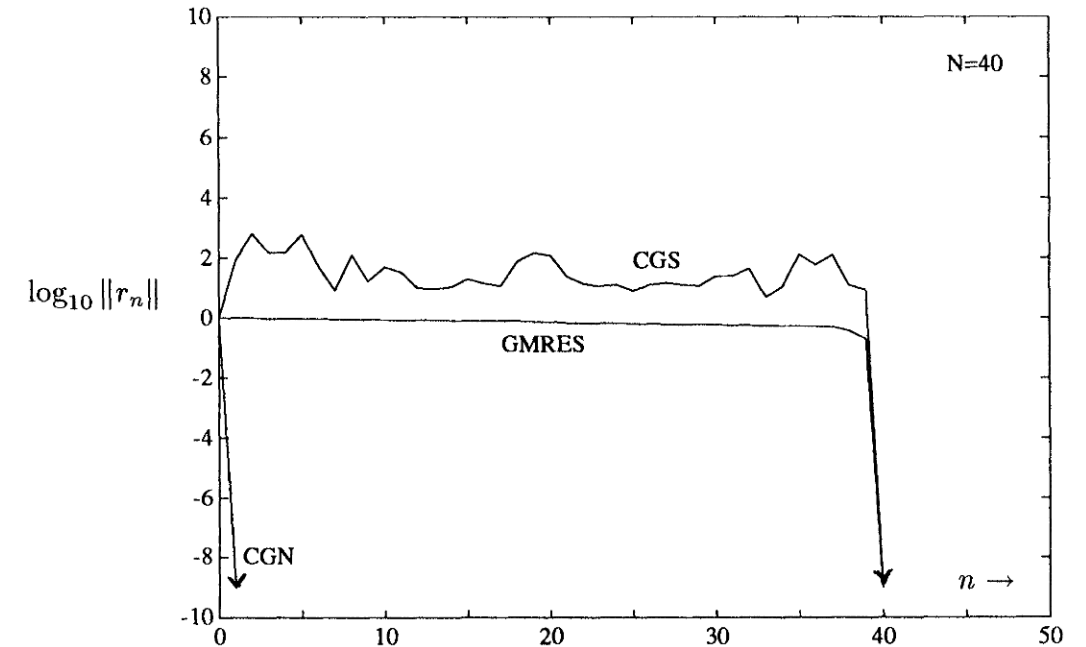
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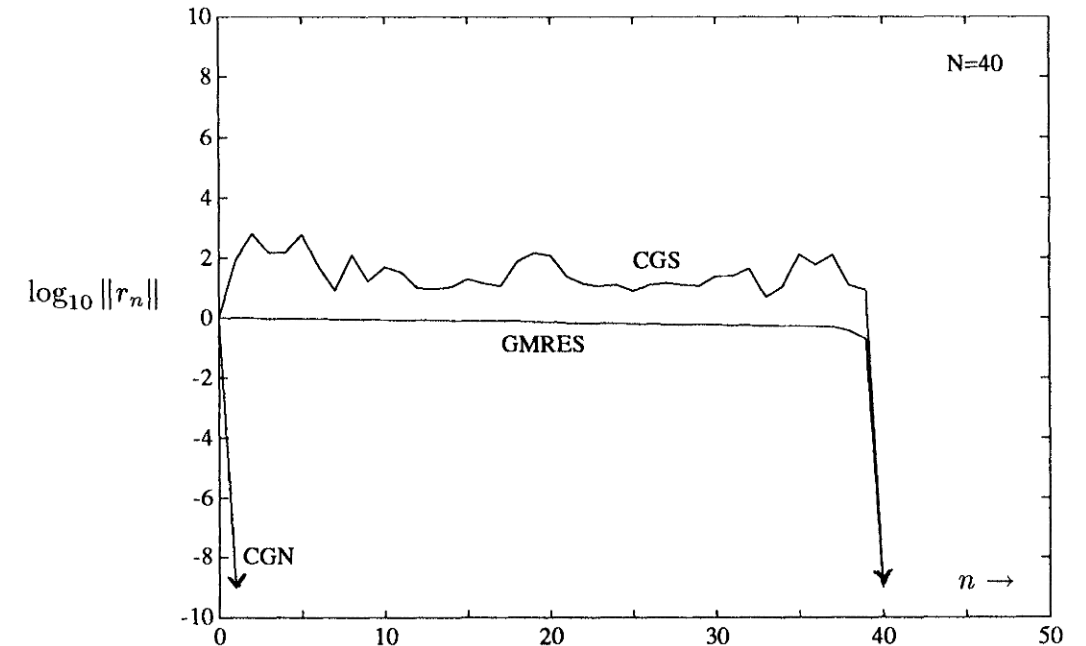
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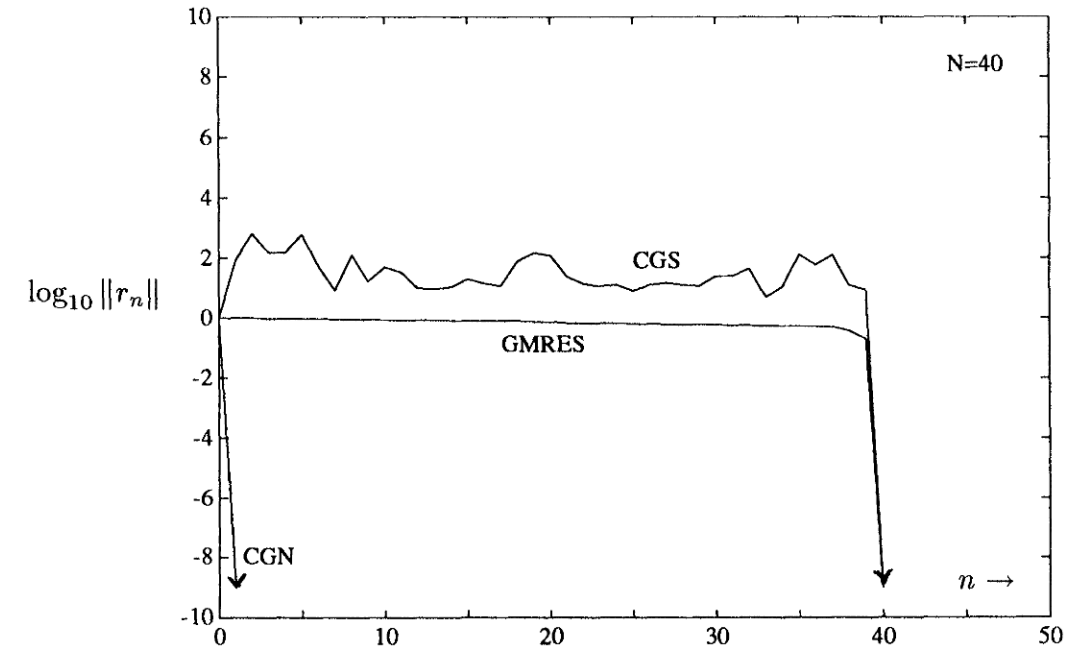
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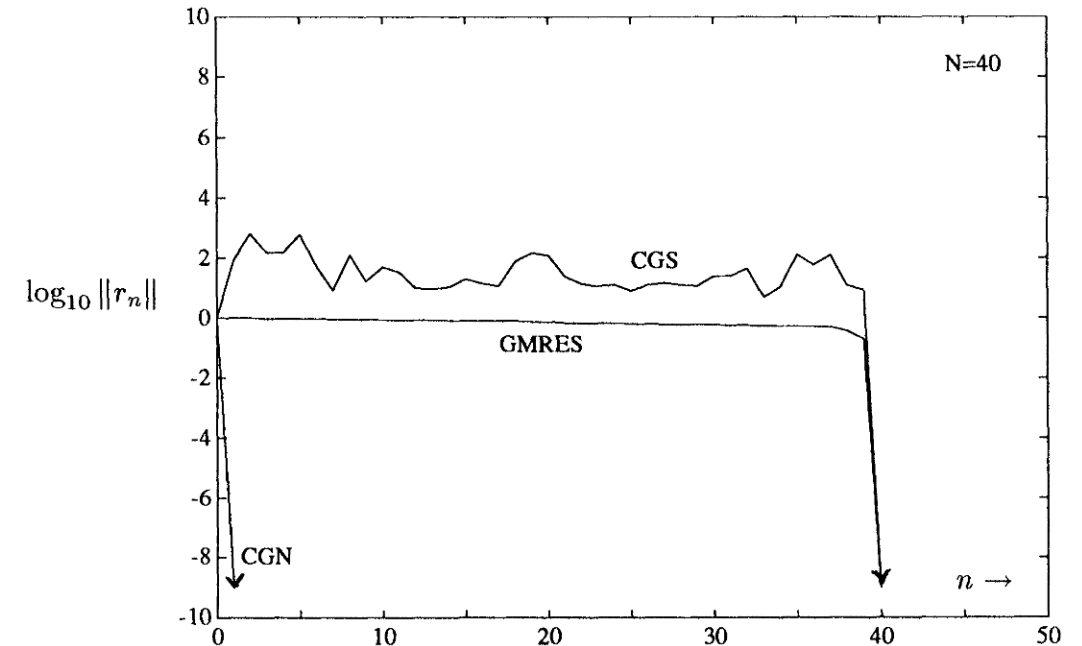
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Equivalent Question for $\kappa = 1$:

How many matvecs $\mathbf{x} \rightarrow A\mathbf{x}$ are needed to approximate the first row of A ?



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From columns to rows



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Hidden Haar Theorem:



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Conclusion

You cannot solve $Ax = b$ with fewer than $n - 1$ matvecs.



Q & A

This is the default photo that came with this powerpoint template.